MIL-I-81537(AS)
26 May 1967
Superseding
WS 4783
2 September 1965

MILITARY SPECIFICATION

IGNITION ELEMENT, ELECTRIC, MARK 13 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 13 MOD 0 ignition element and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Bureau of Naval Weapons

OS 9862	Propellant, Double Base (Nitrocellulose-Nitroglycerine)
MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 1377

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DRAWINGS

Bureau of Naval Weapons

LD 613460	Ignition Element, Electric, MARK 13 MOD 0
LD 613457	Firing Test Fixture, Ignition Element (Output and Electrical Functioning)

PUBLICATIONS

BUSANDA Instruction 4030.17	Preservation, Packaging, Packing and Marking; Navy Policy for
WR-43	Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 13 MOD 0 ignition element covered by this specification shall be manufactured in accordance with LD 613460 and all documents listed thereon.

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3.2 Material. All materials used in the manufacture of the ignition elements shall conform strictly with the specifications referred to on the respective Bureau of Naval Weapons drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 146 ignition elements shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification the loaded spacer charge, Bureau of Naval Weapons Drawing 2302165 of LD 613460, is defined as a primary component. See 4.3.3.

3.4 Cleanliness and condition of parts. All parts shall be free of chips, dirt, grease, rust and all other foreign materials. There shall be no burrs nor other imperfections of manufacture which would harmfully affect either the assembly or the functioning of the product. The cleaning methods used shall not be injurious to any parts nor shall the parts be contaminated by the cleaning agents used.

3.5 Explosive components, volatiles. The primer composition used in the loaded spacer (Bureau of Naval Weapons Drawing 2302165 of LD 613460) shall have a maximum moisture content of 0.5% at time of loading as determined in accordance with 4.7.1. After consolidation of the primer composition, the loaded spacer (Bureau of Naval Weapons Drawing 2302165 of LD 613460) shall be dried at 100° - 120°F for a minimum of 16 hours. After removal of the units from the drying equipment, if more than four hours elapses before their assembly into ignition elements, the units shall be fried at 100° - 120°F for a minimum of four hours immediately prior to their being assembled into an ignition element.

3.6 Insulation resistance. Prior to assembly of bridgewires in the subassembly, the insulation resistance shall not be less than 50 megohms at a relative humidity of less than 80 percent when measured in accordance with 4.7.2.

3.7 Bridgewire.

3.7.1 Bridgewire assembly, continuity. All assemblies shall pass the continuity test of 4.7.3.

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3.8 Igniter assembly, loading. The consistency of the ignition charge and the loading procedure shall be regulated to secure an ignition charge which is free from internal voids while avoiding damage to the bridgewire. The ignition charge shall fill the cavity such that the charge is flush with the top of the cavity and the bridgewire is completely covered. All external surfaces of the igniter assembly shall be free from any of the ignition charge. Where necessary, external surfaces may be cleaned of ignition charge by means of a cloth wet with butyl acetate. The charged igniter assembly shall be dried at a temperature of 100° - 120°F for a minimum of 12 hours. After removal of the units from the drying equipment, if more than 4 hours elapses before their assembly into ignition elements, the units shall be dried at 100° - 120°F for a minimum of 4 hours immediately prior to their being assembled into an ignition element.

3.8.1 Ignition charge. The ignition charge shall be compounded by an approved method which maintains the weight composition of nonvolatile constituents prescribed on Bureau of Naval Weapons Drawing 1558439 of LD 613460. The finished ignition charge when ready for buttering into the igniter assembly shall be a uniformly mixed slurry in which all of the particles are wetted. The consistency shall be such that the liquid will not separate from the mixture when the dish is tilted. Adjustments of the consistency may be obtained by the addition of pure butyl acetate or by evaporation of the solvent. An advisory procedure for preparing and loading the ignition charge is given in 6.5.

3.9 Ignition element inspection.

3.9.1 Visual inspection. The complete ignition element shall be free of the visible defects noted in 4.7.5.

3.10 Bridgewire resistance. The bridgewire resistance of each completely assembled ignition element manufactured, when measured electrically as specified in 4.7.4, shall be not less than 0.08 ohm nor more than 0.14 ohm.

3.11 Tests.

3.11.1 Electric sensitivity.

3.11.1.1 No-fire. The ignition element assembly shall not function when an electric current of 2.00 amperes is applied in accordance with 4.7.6.1. An ignition element which fires is considered defective and is sufficient cause for rejection of the lot represented.

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3.11.1.2 All-fire. The ignition element assembly shall fire when an electric current of 3.00 amperes is applied in accordance with 4.7.6.2. An ignition element which does not fire is considered defective and is sufficient cause for rejection of the lot.

3.11.2 Functioning. Ignition elements from each production lot, selected in accordance with 4.7.7, shall meet the requirements listed below when subjected to the test prescribed in 4.7.7.

3.11.2.1 Response time and output. When assembled in the firing test fixture (LD 613457) and subjected to the firing current prescribed in 4.7.7, the ignition element shall fire in a period of time not to exceed 15 milliseconds. When fired as prescribed in 4.7.7, the ignition element shall cause the puncture of a brass diaphragm 0.008 thick, (Bureau of Naval Weapons Drawing 2302112 of LD 613457) and subsequent thereto, successfully ignite a charge of approximately 0.080 gram of double base propellant, OS 9862, Type I. An ignition element which fails to fire in the prescribed time or to meet the output requirement shall be considered defective and shall cause rejection of the lot represented.

3.12 Workmanship. Ignition elements shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105. Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

4.3 Lots.

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4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 146 ignition elements shall be prepared. One hundred thirty-six of these ignition elements shall be expended in the tests listed below and ten shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.10 of this specification. The contractor shall deliver the ignition elements to the activity designated by the contracting agency for test. Preproduction sample acceptance shall consist of the following tests as prescribed in Table 3 of MIL-D-21625; vibration (12), 40 ft drop (6), temperature and humidity cycling (18), +70°F (50). Number in parentheses denotes number of cartridges allocated for test. No ignition element shall produce results outside the limits set forth in paragraph 3.11.2. In addition, 25 each ignition elements shall be subjected to and meet the requirements of the no-fire and all-fire tests of paragraphs 3.11.1.1 and 3.11.1.2 of this specification. Firings shall be conducted in the firing test fixture LD 613457. If any ignition element fails to meet the requirements specified herein, the preproduction sample shall be rejected. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk. The preproduction sample of ignition elements will not be applied as a part of the quantity specified for delivery by the contract.

4.3.3 Production lot. The suggested production lot size is 1201 minimum and 3200 maximum ignition elements. Ignition elements required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single batch shall be used in a production lot of ignition elements. One primary component batch may be used in more than one ignition element production lot. Ignition elements selected for the functional test shall be delivered to the activity designated by the contracting agency.

4.4 Gages. The contractor shall provide himself with gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C), and

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major characteristics, by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

4.6 Production and acceptance inspection. Production and acceptance inspection shall consist of verification of the characteristics classified on the drawings of LD 613460 and the following tests and examinations.

Item	Nature of Test	Requirement	Test	Classification (WR-43)
Primary Component	Single Lot Verification	3.3	4.3.3	(M101)
Primer Composition	Moisture Content	3.5	4.7.1	(C1)
Primer Composition	Drying Verification	3.5	---	(M102)
Inner Cup and Electrode Assembly	Insulation Resistance	3.6	4.7.2	(M103)
Bridgewire Assembly	Continuity	3.7.1	4.7.3	(C2)
Igniter Assembly	Loading Verification	3.8	---	(C3)
Igniter Assembly	Drying Verification	3.8	---	(M104)
Ignition Element	Visual Inspection	3.9.1	4.7.5	(M105)
Ignition Element	Bridgewire Resistance	3.10	4.7.4	(C4)
Ignition Element	No-fire	3.11.1.1	4.7.6.1	(C5)
Ignition Element	All-fire	3.11.1.2	4.7.6.2	(C6)
Ignition Element	Functioning	3.11.2	4.7.7	(C7)

4.6.1 Packing, packaging and marking. (M106) The inspector shall ascertain that the packing and packaging of the electric ignition elements and the container markings conform to Section 5 of this document.

4.7 Tests.

4.7.1 Determining moisture content. Transfer approximately 2.0 grams of the primer composition from the representative sample to a tared weighing dish, weigh to the nearest 0.1 milligram, calculate the weight of the sample, transfer weighing dish and sample to a vacuum drying oven, dry at +71°C ± 2°C and 28 ± .2 inches of mercury vacuum until constant weight is obtained. The loss of weight is calculated as the percentage of moisture in the original sample.

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4.7.2 Insulation resistance. Prior to assembly of the bridgewires, the insulation resistance shall be checked with an applied voltage of 500 ± 25 volts d.c. This measurement shall be made between the electrode and the inner cup. Units which fail to meet the requirements of 3.6 shall be rejected.

4.7.3 Bridgewire assembly, continuity. The assembly shall be thoroughly cleaned to remove flux and contamination. Subsequent to cleaning, the bridgewires shall be flashed by application of an electric current to assure continuity of both bridges. The current shall be regulated to produce a visible red glow of the bridges. An EMF of approximately 1.5 volts a.c. or d.c. may be used with a suitable resistor to adapt the source characteristics to the bridge resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to glow in both bridges shall be rejected. The faulty assemblies may be reworked by replacement of the bridges.

4.7.4 Bridgewire resistance. Conformance with the bridge circuit resistance of 3.10 shall be ascertained by means of a test circuit which limits the bridge current to 25 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable bridge which is known to be accurate within 1% in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirements of 3.10 shall be rejected. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit.

4.7.5 Visual inspection. Examine each ignition element and reject for chips, burrs, cracks, split or perforated closure disc or for any other imperfections which would harmfully affect either assembly or functioning.

4.7.6 Electric sensitivity tests.

4.7.6.1 No-fire test. The number of ignition elements selected from each production lot for the no-fire test shall be in accordance with the requirements of Table I of MIL-STD-105, Inspection Level I. Each ignition element shall be conditioned at +70°F ± 5°F for a period 1 hour longer than that required to reach thermal equilibrium and then tested in the firing test fixture, LD 613457. The firing circuit shall contain a resistor in series sufficient to regulate the current to $2.0^{+0.100}_{-0.000}$ amperes measured to within an accuracy of 1%. This current shall be applied to the ignition element for a period of not less than 1 minute nor more than 1.1 minutes. An ignition element which fires shall be considered defective and shall cause rejection of the lot represented. If the ignition elements pass this test they shall then be subjected to the test of paragraph 4.7.6.2 below.

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4.7.6.2 All-fire test. The ignition element shall be conditioned at +70°F ± 5°F for a period 1 hour longer than that required to reach thermal equilibrium and then fired in the firing test fixture, LD 613457. The firing circuit shall contain a resistor in series sufficient to regulate the current to 3.0 $^{+0.100}_{-0.000}$ amperes measured to within an accuracy of 1%. Any ignition element which fails to fire within 0.100 milliseconds after current application shall be considered defective and shall cause rejection of the lot represented.

4.7.7 Functional test. The number of ignition elements selected from each production lot for the functional test shall be in accordance with the requirements of Table I of MIL-STD-105, Inspection Level II. The ignition elements shall be conditioned at +70°F ± 5°F for a period 1 hour longer than that required to reach thermal equilibrium and fired in the firing test fixture, LD 613457. In addition to the sample of ignition elements for the tests of 4.7.6.1, 4.7.6.2 and 4.7.7, ten additional ignition elements shall be furnished from the production lot for investigative purposes.

4.7.7.1 Criteria of acceptability.

4.7.7.1.1 Response time and output. Each ignition element shall be mounted in the firing test fixture, LD 613457, and the firing circuit shall contain a resistor in series sufficient to regulate the current to 5.0 amperes measured to within an accuracy of 1%. The firing interval between the application of current to the ignition element and the detection of explosive discharge (firing) shall be measured using an electronic timer or equally responsive instrument accurate to 1.0 percent. The timer is activated by an applied pulse to the ignition element and is stopped when the explosive discharge severs a graphite or wire link in the pulse circuit. The 0".008 thick brass diaphragm, Bureau of Naval Weapons Drawing 2302112 of LD 613457, used to measure the output shall be located 2".814 ± 0".015 from the face of the ignition element's closure disc when assembled in the test fixture. The 0.080 gram of double base propellant, OS 9862 Type I, shall be loaded in the firing test fixture as prescribed in LD 613457. Any ignition element which fails to meet the requirements of 3.11.2 shall be considered defective and shall cause rejection of the lot represented.

4.8 Retest. There shall be no retests.

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4.8.1 Test failure. If test failure is attributable to an assignable cause, excluding the test ignition elements, original test results should be discarded and the complete test reconducted.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, ignition elements which are intended for expenditure in preproduction sample testing and for production lot acceptance tests or shipment to the point of assembly into a cartridge shall be packaged in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 71-78. For a definition of Level C packaging, see 6.4.

5.2 Packing.

5.2.1 Level A. Not applicable.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, ignition elements, which are intended for expenditure in preproduction sample testing and for production lot acceptance tests or shipment to the point of assembly into a cartridge and are packaged in accordance with 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78. For a definition of Level C packing, see 6.4.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 71-78.

5.3.2 Normal markings. In addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

6.1 Intended use. This ignition element was originally intended for use in the MARK 17 MOD 0 impulse cartridge.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number and date of this specification;

b. whether preproduction inspection is required and if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List) which shall be attached to and made a part of the contract or order. NAVWEPS Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. The definition of Level C packaging and packing given below was extracted from BUSANDA Instruction 4030.17.

6.4.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

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6.4.2 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 Explosives safety precautions.

6.5.1 Applicability. The loading, assembly and handling of the explosives, subassemblies and the finished items covered by this specification involve hazardous operations and therefore require suitable explosive safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed below are mandatory. These same minimum provisions are recommended for observance by a contractor or manufacturer that explosively loads the item covered in this specification in order to fulfill his own safety responsibility. Other warnings and precautions, pertaining to the operational effectiveness or safety of the specified item during its use (for example, during its handling, launching or firing), are included in the detailed technical requirements of the specification. Use of this specification shall not be construed to replace other safety regulations of governmental agencies or to relieve the contractor or manufacturer of responsibility for the safety of his operations.

6.5.2 Personnel and equipment. Proper care must be exercised at all times to protect personnel from accidents, fires or explosions and to limit damage to equipment and loading areas. In connection with the above, the precautionary measures in the following paragraphs should be observed. (See also paragraph 6.5.4, Static and electrical and other applicable parts of these safety precautions.)

6.5.2.1 Explosives operations shall be kept under strict supervision. Periodic inspections shall be employed to assure compliance with established safety regulations.

6.5.2.2 Safe equipment and methods should be utilized for handling and transporting explosives and loaded parts.

6.5.2.3 Where required, remote control-barricaded-handling equipment shall be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

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6.5.2.4 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.2.5 Maintain equipment such as presses, dies, crimpers, blenders, mixers, sifters, conveyors, and the like in safe operating condition.

6.5.2.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.2.7 During static tests, propulsion units such as rocket motors shall be restrained against full thrust to prevent flight if ignited, for example, during test of igniter circuit. Also, means shall be provided to harmlessly vent flames and exhaust gases.

6.5.3 Operating procedures. All loading operations shall be conducted in full accordance with a formalized operating procedure which shall include the following:

6.5.3.1 Keep only minimum, limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.3.2 Allow no fires, exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.3.3 Appropriate safety instructions for specific operations shall be posted and personnel instructed as to the hazards present and precautions involved.

6.5.3.4 The number of working personnel and transients permitted in each work space shall be established and posted in such areas. The numbers shall be the minimum number required to accomplish necessary operations.

6.5.3.5 All loading operations should be conducted in a neat and orderly manner.

6.5.3.6 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings.

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6.5.3.7 Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from, floors, walls, ceiling, ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

6.5.3.8 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.3.9 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.3.10 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges. (See paragraph 6.5.4.)

6.5.4 Static and electrical.

6.5.4.1 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges shall be employed. Fabrics, such as silk and nylon, which promote static generation shall be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type of electric primers, detonators and squibs. The latter types of items shall have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations, such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast; also electrical shielding from induced electric currents originated by sources such as lightning, static, electro-magnetic radiations, etc. Where necessary for safety, humidity of

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work rooms should be appropriately increased, as required to lessen electrostatic effects, but without inducing excessive moisture absorption by any of the components of the item being processed.

6.5.4.2 Protect all explosive operations from effects of electric current originating in equipment (such as soldering irons or guns, heaters, switches, wiring, motors, lights, test instruments, etc.) by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or by completing an electric circuit through an electric primer, detonator or squib. For example, circuits may be inadvertently completed from a defective electric soldering iron through a grounded contact. All electric type of primers, detonators or squibs provided with wire leads shall have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Removable short circuiting clips, or other devices should be employed during manufacturing operations involving items such as electric primers, detonators or squibs.

6.5.4.3 Where necessary, protect personnel and operations from electromagnetic energy. In certain work with very sensitive explosives or items, a shielded environment free from significant electromagnetic energy should be provided.

6.5.5 Operating facility and storage regulations.

6.5.5.1 All explosives including partially loaded items should be stored in suitable storage magazines located in accordance with the minimum distances prescribed in the American Table of Distances (ATD). While the explosive or loaded items are in process, stored in loading rooms, or in ready or service magazines, they should be located in accordance with Intraplant Distances set forth in the ATD. For Navy managed or contractor managed-Navy owned-explosives loading plants, the Armed Services Explosives Safety Board standards approved by the Department of Defense, the Secretary of the Navy and the Naval Air Systems Command covering quantity-distance regulations for explosives apply.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE :	1. DOCUMENT NUMBER MIL-I-81537(AS)	2. DOCUMENT DATE (YYMMDD) 670526

3. DOCUMENT TITLE

IGNITION ELEMENT, ELECTRIC. MARK 13 MOD 0

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-1963 (2) DSN 354-1963
c. ADDRESS (*Include Zip Code*) Engineering Documentation (Code 4230) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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